Highlights

Independent Real-Time Verification of Manual Valve Position as an Additional Protection Layer Against Misrouting During Product Transfers: A Monte Carlo Barrier-Effectiveness Assessment

- Independent position sensing cuts modeled undetected misrouting 14-fold
- Over assumed priors the reduction factor is 9.7 (5th–95th percentile 4.4–28)
- Operator response to alarms is the weakest link: 90% response gives 5.6-fold
- An 8% alarm tolerance stays near-optimal once alarm fatigue is included
- Exact binomial acceptance criteria are derived for a pilot of 1000 valves